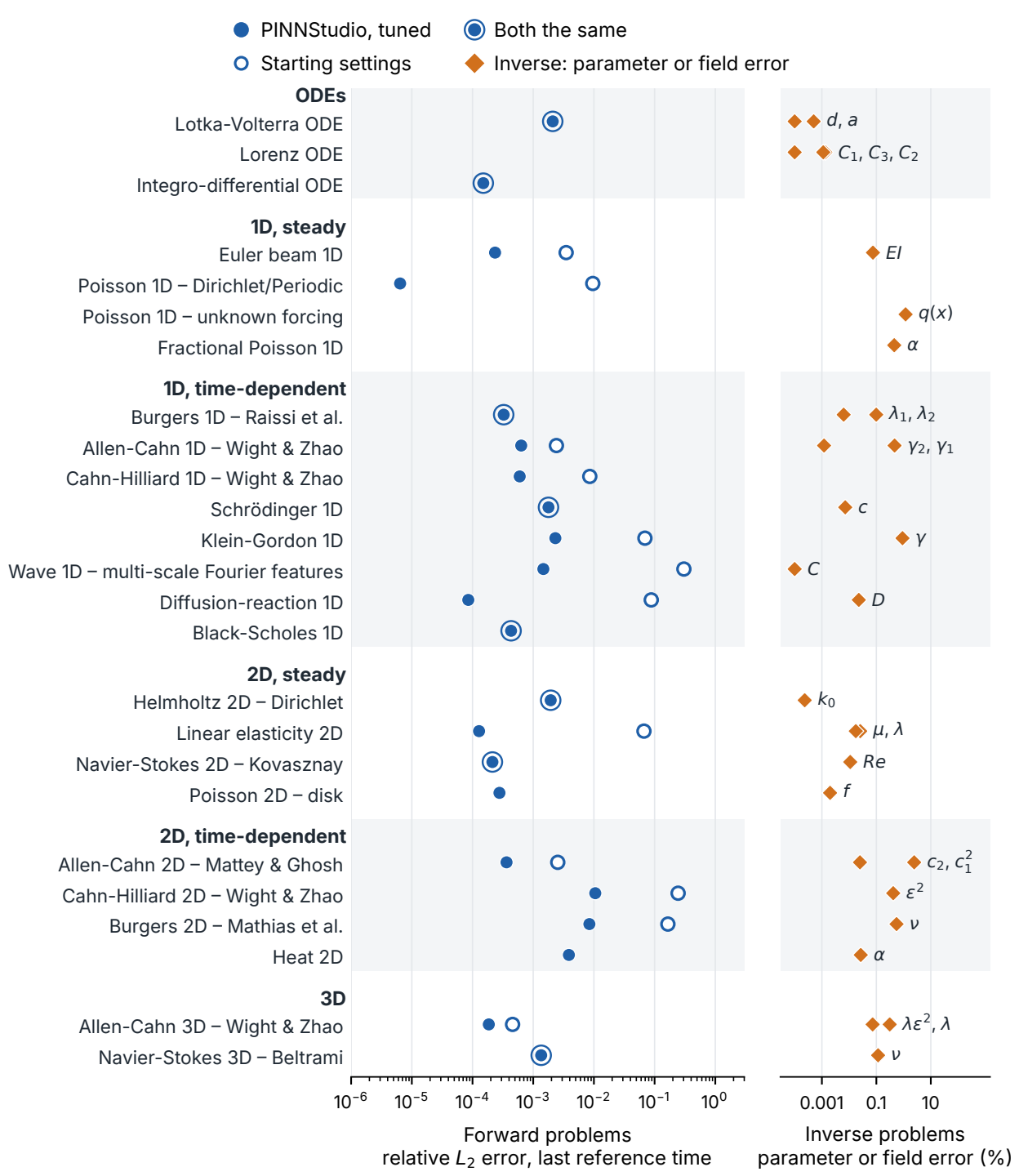
PINNStudio, tuned
Starting settings
Both the same
Inverse: parameter or field error
ODEs
Lotka-Volterra ODE
Lorenz ODE
Integro-differential ODE
1D, steady
Euler beam 1D
Poisson 1D – Dirichlet/Periodic
Poisson 1D – unknown forcing
Fractional Poisson 1D
1D, time-dependent
Burgers 1D – Raissi et al.
Allen-Cahn 1D – Wight & Zhao
Cahn-Hilliard 1D – Wight & Zhao
Schrödinger 1D
Klein-Gordon 1D
Wave 1D – multi-scale Fourier features
Diffusion-reaction 1D
Black-Scholes 1D
2D, steady
Helmholtz 2D – Dirichlet
Linear elasticity 2D
Navier-Stokes 2D – Kovasznay
Poisson 2D – disk
2D, time-dependent
Allen-Cahn 2D – Mattey & Ghosh
Cahn-Hilliard 2D – Wight & Zhao
Burgers 2D – Mathias et al.
Heat 2D
3D
Allen-Cahn 3D – Wight & Zhao
Navier-Stokes 3D – Beltrami
d, a
C1, C3, C2
EI
q(x)
α
λ1, λ2
γ2, γ1
c
γ
C
D
k0
μ, λ
Re
f
c2, c1^2
ε^2
ν
α
λε^2, λ
ν
10^−6 10^−5 10^−4 10^−3 10^−2 10^−1 10^0
Forward problems
relative L2 error, last reference time
0.001 0.1 10
Inverse problems
parameter or field error (%)